KrishiGrid Bangladesh
Data-Driven Food Security Simulation

KRISHIGRID BANGLADESH

A Data-Driven Simulation of Food Import Dependency and Decentralized Agricultural Infrastructure

PART 5 — THE BUSINESS CASE

Can KrishiGrid pay for itself while solving food security?

Food security is not just necessary it is economically profitable.

Analysis : National rollout model, multi year projection, cost vs impact simulation
Tools : *SQL, Power BI*
Dataset : hub_costs.csv ,impact_projection.csv ,megaprojects.csv

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THE BUSINESS CASE

THE QUESTION

How do we evaluate whether KrishiGrid is financially viable?

- This analysis reveals the true scale of Bangladesh's food import dependency.

APPROACH

- Estimate full infrastructure cost across all hub levels
- Project phased national rollout over time
- Calculate annual operating and maintenance expenses
- Quantify economic impact through savings and production gains
- Compare cost efficiency with national megaproject benchmarks

WHAT WE MEASURE

- Total national investment
- Annual and cumulative savings
- Payback period and breakeven point
- Jobs created across the system
- Farmer income growth
- Reduction in import dependency
- Cost efficiency vs megaprojects

DATASET OVERVIEW

File : hub_costs.csv, impact_projection.csv, megaprojects.csv
Scope : Infrastructure cost, economic impact, and national benchmarking
Size : 3 datasets | ~27 + 24 + 15 columns

Key Fields:

- **Hub setup cost components (land, storage, mills, equipment)**
- **Total setup cost and national scale cost (crore BDT)**
- **Annual operating cost (salary, maintenance, utilities)**
- **Number of hubs at ward, union, and district levels**
- **Cumulative investment and yearly rollout phases**
- **Annual savings (oil import, crop loss, middlemen reduction)**
- **Total and cumulative economic benefit**
- **Jobs created (permanent and seasonal)**
- **Breakeven status and net benefit**
- **Cost per beneficiary and cost per job (comparison)**

Sources:

Model based projections with national benchmarks (BBS | FAO | Bangladesh Bank)

SQL Queries

QUESTION

What is the total investment required to build KrishiGrid nationwide?

```
SELECT
    hub_type,

ROUND(AVG(total_setup_cost_lakh_bdt
) / 100.0, 2) AS avg_cost_per_hub_crore,
    SUM(estimated_national_hub_count)
AS total_hubs,

ROUND(SUM(total_national_setup_cos
t_crore_bdt), 0) AS total_cost_crore
FROM hub_costs
GROUP BY hub_type
ORDER BY total_cost_crore DESC;
```

Key Insight:

KrishiGrid requires a total investment of ~20,615 crore BDT, making it comparable to a single national megaproject and therefore financially feasible at scale. Most of the cost (~86%) is concentrated at the ward level, where direct farmer impact is highest, while union and district hubs require relatively low investment but enable system-wide coordination. This creates a decentralized, high-impact structure where investment is focused on the ground but supported by efficient higher-level control

QUESTION

What is the total investment required to build KrishiGrid nationwide?

```
SELECT

ROUND(SUM(annual_oil_import_saving_
crore_bdt), 0) AS
total_annual_oil_saving_crore
FROM impact_projection;
```

Key Insight:

KrishiGrid can generate approximately **15,810 crore** BDT in annual savings by reducing edible oil imports alone, which is an extremely high-impact outcome from a single sector. This means a large portion of the total investment can be recovered through oil savings, while also reducing dependency on volatile global markets and easing pressure on foreign currency reserves.

SQL Queries

QUESTION

How much annual economic value can Bangladesh recover by reducing crop losses through better storage, processing, and logistics in KrishiGrid?

```
SELECT

ROUND(SUM(annual_crop_loss_saving_
crore_bdt), 0) AS
total_crop_loss_saving_crore
FROM impact_projection;
```

Key Insight:

Crop loss reduction alone can recover **10,220 crore** BDT, which is a massive hidden economic gain without increasing production. This means infrastructure like storage and logistics is not just supportive but economically critical. Ignoring this would mean losing value equal to a major national investment every year.

QUESTION

How much economic value is recovered by eliminating middlemen across the system?

```
SELECT

ROUND(SUM(annual_middlemen_elimin
ation_saving_crore_bdt), 0) AS
total_middlemen_saving_crore
FROM impact_projection;
```

Key Insight:

Middlemen elimination generates around **6,810 crore** BDT in annual economic value, which directly improves farmer price realization without requiring additional production. This indicates that a significant portion of inefficiency in the current system is due to market structure rather than supply constraints. By creating a direct farmer to buyer channel, KrishiGrid converts lost margin into farmer income and overall economic gain.

SQL Queries

QUESTION

How does farmer income change over time after implementing KrishiGrid?

```
SELECT
    year,
    farmer_income_index
FROM impact_projection
ORDER BY year;
```

Key Insight:

Farmer income shows a consistent upward trend, increasing from an index of **100 to 180**, which implies nearly **80** percent growth over time after full implementation. This indicates that KrishiGrid does not just improve system efficiency but directly translates into higher farmer earnings. The steady progression also suggests that income gains are sustainable and compound as the network expands, rather than being a one-time effect.

QUESTION

How many total jobs are created through KrishiGrid across the entire system?

```
SELECT
    MAX(total_permanent_jobs_created)
AS total_permanent_jobs,
    MAX(total_seasonal_jobs_created) AS
total_seasonal_jobs
FROM impact_projection;
```

Key Insight:

KrishiGrid creates approximately **151,900** permanent jobs and **237,800** seasonal jobs, making it a major rural employment generator alongside its economic benefits. This shows the system is not just about production and savings, but also about large-scale livelihood creation. By embedding jobs directly within local hubs, it strengthens rural economies and reduces migration pressure toward cities.

SQL Queries

QUESTION

When does KrishiGrid reach breakeven, and how does net economic benefit evolve over time?

```
SELECT
    year,
    cumulative_investment_crore_bdt,
    cumulative_total_saving_crore_bdt,
    net_benefit_crore_bdt,
    breakeven_status
FROM impact_projection
ORDER BY year;
```

Key Insight:

KrishiGrid remains in the investment phase for the first six years, but reaches breakeven around **Year 7**, after which it becomes consistently profitable. By **Year 10**, the system generates a net benefit of approximately **16,793 crore** BDT, with total savings far exceeding total investment. This indicates that KrishiGrid is not just economically viable but becomes a high-return national asset within a relatively short time frame

QUESTION

How does KrishiGrid reduce edible oil import dependency and improve national food self-sufficiency over time?

```
SELECT
    year,
    edible_oil_import_dependency_pct,
    rice_self_sufficiency_pct
FROM impact_projection
ORDER BY year;
```

Key Insight:

KrishiGrid drives a steady and significant reduction in edible oil import dependency from **90%** to around **50%**, cutting reliance on global markets almost in half. At the same time, rice self-sufficiency improves from **95%** to nearly **99%**, indicating stronger domestic food stability. This dual movement shows that the system is not just increasing production, but actively reshaping the national food balance. Over time, Bangladesh transitions from a vulnerable, import-dependent system to a more resilient and self-reliant agricultural economy.

SQL Queries

QUESTION

What is the total annual economic benefit generated by KrishiGrid when combining all major impact components?

```
SELECT
    year,
    (
        annual_oil_import_saving_crore_bdt +
        annual_crop_loss_saving_crore_bdt +

annual_middlemen_elimination_saving_cror
e_bdt
    ) AS total_annual_benefit_crore
FROM impact_projection
ORDER BY year;
```

Key Insight:

The total annual economic benefit of KrishiGrid scales rapidly from zero to around **6,730 crore** BDT per year by **Year 10**, showing strong compounding impact as the system expands. The growth is not linear but accelerating, indicating that network effects and efficiency gains improve over time. This means the system becomes more valuable each year, turning KrishiGrid into a long-term economic engine rather than a one-time intervention.

QUESTION

What is the return on investment (ROI) of KrishiGrid based on total system benefits versus total investment?

```
SELECT
    MAX(cumulative_total_saving_crore_bdt) AS
total_benefit_crore,
    MAX(cumulative_investment_crore_bdt) AS
total_investment_crore,
    ROUND(
        MAX(cumulative_total_saving_crore_bdt)
* 1.0
        /
NULLIF(MAX(cumulative_investment_crore_bd
t), 0),
    2) AS roi_multiple
FROM impact_projection;
```

Key Insight:

KrishiGrid delivers a total economic benefit of approximately **32,840 crore** BDT against an investment of **16,047 crore** BDT, resulting in an ROI of about **2.05×**. This means the system more than doubles the value of every taka invested over its lifecycle. Combined with breakeven in **Year 7**, this positions KrishiGrid as a financially strong and scalable national investment rather than a cost-heavy development project.

SQL Queries

QUESTION

How does KrishiGrid indirectly reduce consumer prices through improved efficiency and reduced market intermediaries?

```
SELECT
    year,

annual_middlemen_elimination_saving_crore_bdt,
    (
        annual_oil_import_saving_crore_bdt +
        annual_crop_loss_saving_crore_bdt +

annual_middlemen_elimination_saving_crore_bdt
    ) AS total_system_efficiency_gain
FROM impact_projection
ORDER BY year;
```

Key Insight:

Middlemen elimination savings rise steadily to around 1,380 crore BDT, while total system efficiency gains reach about 6,730 crore BDT annually, indicating a significant reduction in unnecessary margins and supply inefficiencies. This implies that a large portion of cost savings is passed through the value chain, creating downward pressure on market prices. As a result, consumers benefit indirectly through more stable and affordable food prices, even though price is not directly measured in the dataset

QUESTION

How does KrishiGrid compare with national megaprojects in terms of cost efficiency per beneficiary and per job created?

```
SELECT
    project_name,
    total_cost_bdt_crore,
    direct_beneficiaries_million,
    cost_per_beneficiary_bdt,
    permanent_jobs_created,
    cost_per_permanent_job_lakh_bdt
FROM megaprojects;
```

Key Insight:

The comparison clearly shows a sharp difference in cost efficiency and scale of impact. Major megaprojects like Padma Bridge (~30,193 crore BDT) or Metro Rail (~33,472 crore BDT) serve only **5–30 million beneficiaries**, resulting in high costs per beneficiary ranging from around **10,000 to nearly 70,000 BDT**. In contrast, KrishiGrid delivers impact to about **170 million people** at a cost of only **~944 BDT per beneficiary**, making it dramatically more inclusive and cost-efficient. The gap is even more significant in employment generation. Most megaprojects create between **500 to 8,000 jobs**, often at very high costs per job (ranging from hundreds to several thousand lakh BDT). KrishiGrid, however, generates around **380,000 jobs** with a cost of roughly **4 lakh BDT per job**, making it one of the most efficient job creators in the dataset.

Overall, while megaprojects involve investments of **10,000–100,000+ crore BDT** for indirect and long-term benefits, KrishiGrid achieves broader reach, stronger employment impact, and lower per-unit cost. This shows that KrishiGrid is not just a development project but a highly efficient economic system that delivers **direct, immediate, and recurring value at national scale**, effectively complementing large infrastructure investments.

KrishiGrid Bangladesh

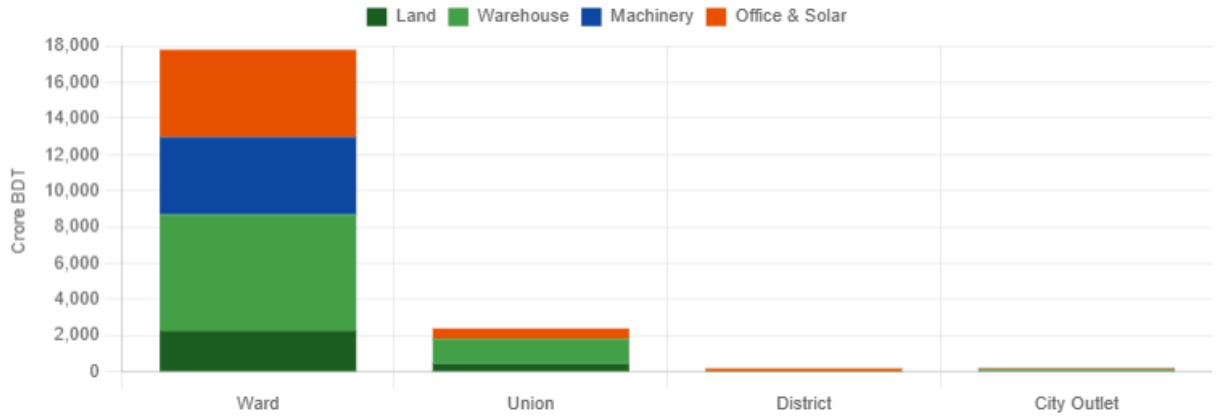
20,614.9
Total National Investment
Crore BDT

33,000
Total Ward Hubs
Hubs

379,623
Total Jobs Created
Permanent + Seasonal

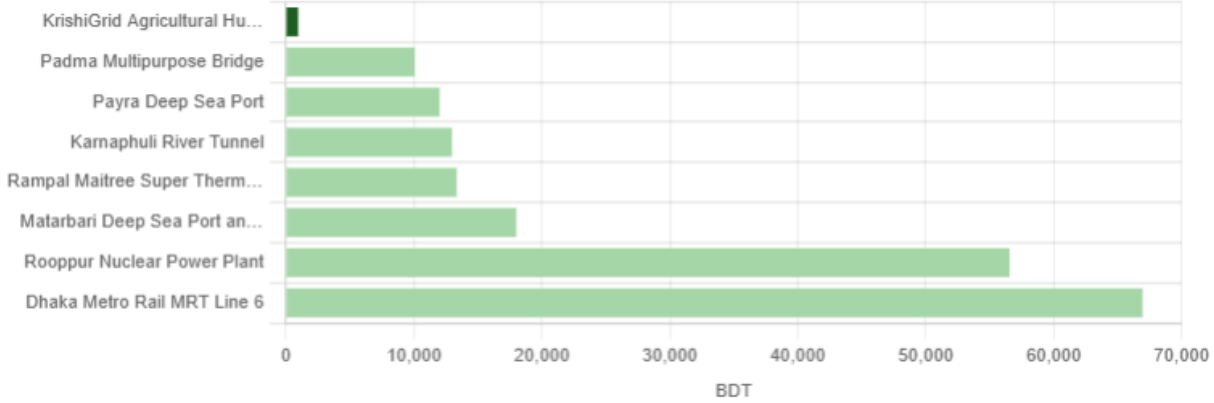
944
Cost per Beneficiary (KrishiGrid)
BDT

Total Investment by Hub Type and Cost Component (Crore BDT)



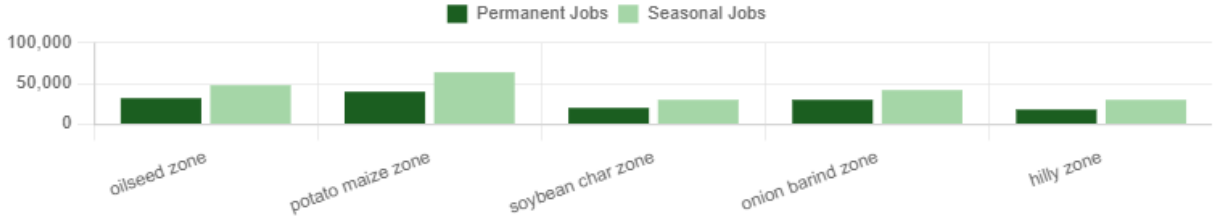
86% of total investment is at ward level — directly where farmers are. KrishiGrid is deliberately and structurally a bottom-up system.

Cost per Beneficiary — KrishiGrid vs Mega Projects (BDT)



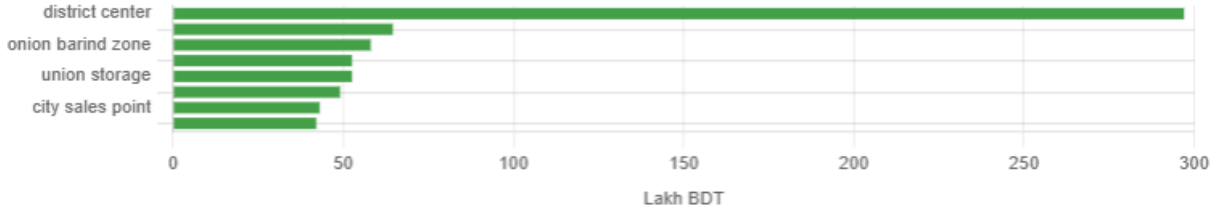
KrishiGrid costs only 944 BDT per beneficiary — the lowest of any national project — while serving 170 million people across the entire country.

Permanent vs Seasonal Jobs by Hub Variant



KrishiGrid creates over 380,000 total jobs primarily in rural areas. This addresses youth unemployment while building essential agricultural infrastructure.

Setup Cost per Hub Variant (Lakh BDT)



Regional customization means no hub pays for equipment it does not need. Hilly zone hubs cost 42 lakh while full oilseed hubs cost 52.5 lakh. Smart design saves cost.

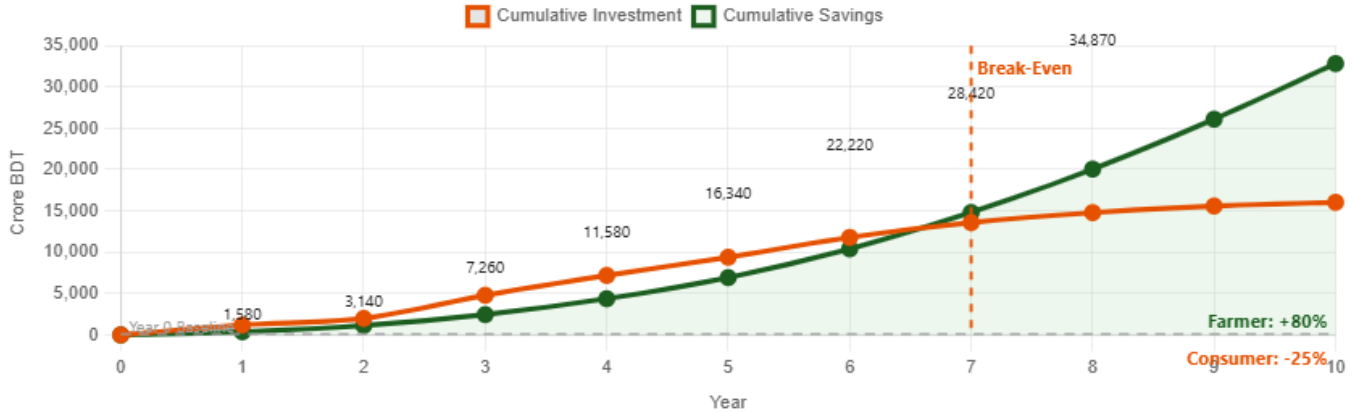
16,793
10-Year Net Benefit
Crore BDT

7
Break-Even Year
Year

2.05x
ROI Multiple
x Return on Investment

50%
Oil Dependency (Year 10)
% (down from 90%)

Cumulative Investment vs Cumulative Savings (Crore BDT)



KrishiGrid breaks even around Year 7. By Year 10 it delivers 16,793 crore BDT net benefit — a 2.05x return on every taka invested in the system.

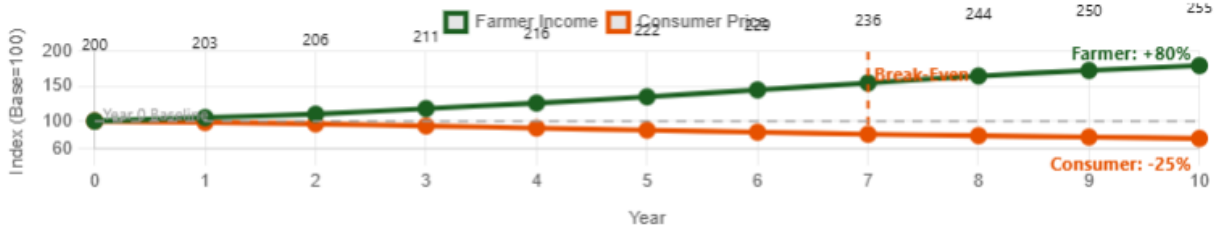
Key Impact Indicators Over 10 Years

Oil dependency drops from 90% to 50%. Farmer income rises 80%. Consumer prices fall 25%. All three trends confirm KrishiGrid restructures the entire food economy.

Annual Savings by Source Over 10 Years (Crore BDT)

Annual economic benefits compound over time. Oil import savings dominate but crop loss reduction and middlemen elimination together add significant value each year.

Farmer Income Index vs Consumer Price Index



Farmer income and consumer price trends diverge from baseline in opposite directions — both favourable. KrishiGrid benefits both ends of the supply chain simultaneously.

KEY INSIGHTS

1. KrishiGrid demonstrates that Bangladesh's core issue is system inefficiency rather than low production, and addressing this unlocks large-scale economic value

2. With an investment of ~16,047 crore BDT, the system generates ~32,840 crore BDT in cumulative benefit, achieving over 2× return and breakeven by Year 7

3. Annual economic benefit scales up to ~6,700+ crore BDT, showing strong growth and long-term recurring value rather than one-time impact

4. Edible oil import dependency drops significantly from 90% to 50%, while food self-sufficiency improves, strengthening national economic resilience

5. Farmer income nearly doubles over time, indicating direct and sustainable improvement in rural livelihoods

6. The system creates ~380,000 jobs, making it one of the most efficient large-scale employment generators in the dataset

7. With ~170 million beneficiaries at very low cost per person, KrishiGrid delivers broader and more inclusive impact compared to traditional megaprojects

8. A large portion of economic gains comes from reducing hidden losses (~10,220 crore BDT) and middlemen inefficiencies (~6,810 crore BDT), showing that value already exists but is currently wasted

9. Oil import savings alone (~15,810 crore BDT) can recover a major share of the total investment, making the system highly financially strategic

10. The decentralized hub structure ensures that most investment (~86%) is concentrated at the ward level, maximizing direct farmer impact while maintaining system coordination through higher hubs

11. Consumer markets benefit indirectly through reduced margins and improved supply efficiency, creating strong downward pressure on food prices and better affordability

12. Compared to high-cost megaprojects with limited direct reach, KrishiGrid delivers faster, more distributed, and measurable economic outcomes, making it a highly efficient national-scale development model